FPGA Lab – Input/Output Subsystem

In this part of the practical sessions, we will study two ways a CPU interacts with the Input/Output subsystem: polling and Interrupt-driven. During the lab session we will use the hardware platform generated in the previous lab, and write software to drive the Universal Asynchronous Receiver/Transmitter (UART) in polling mode. Interrupt driven input/output is left for homework.

Objectives

After completing this lab session you will be able to:

1. Describe the purpose and operating principles of a UART.
2. Explain the difference between polled and interrupt-driven IO.
3. Navigate the relevant AMD documentation to find information for writing a minimal driver.
4. Develop a minimal C application that initialises the UART peripheral and echoes received characters to the **USER2 Vitis console**.

Implementation guidelines during the Lab

Create an empty Vitis application project (Template: Empty Application (C)) and copy the accompanying `main.c` file from the `src` directory of the `polled_io` application project.

The operation of the UART is controlled fully by four memory-mapped registers. Some are used for configuration, others to communicate the status, and lastly some are used for the actual operation. Use the product guide [2] to understand what values these registers (page 16) should contain, then

1. Check whether the UART is enabled by default, and if necessary update it's configuration.
2. Complete the `print(char *s)` function to monitor the UART status and transmit the characters one-by-one.

Note: The IP documentation in [2] can be obtained directly within Vivado by right-clicking the IP block in the block diagram and navigating the documentaion sub-menu.

After verifying that your code can transmit characters, use polling to monitor the UART receiver and echo (transmit) any characters back to the standard output.

References

1. MicroBlaze Processor Reference Guide (UG984)
2. MicroBlaze Debug Module LogiCORE IP Product Guide (PG115)